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Anomaly detection based on traffic packets in cloud environment

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***Abstract*—Because the network traffic data is large and the flow rate is high, the abnormal traffic can provide support for the host vulnerability warning, which makes the real-time detection of abnormal network traffic become a key problem. This paper proposes an online anomaly detection model. Firstly, based on the encrypted traffic features of network traffic header, a threshold is set for the feature quantity and the extraction time of each stream to ensure the timeliness of feature extraction. On this basis, an anomaly classification model is constructed based on the classical LeNet-5. Finally, a simulation experiment was carried out to evaluate the anomaly detection accuracy of the above anomaly detection model. The experimental results showed that a small number of header messages could achieve a high anomaly detection accuracy, which provided certain proof for the feasibility of online anomaly detection, also provided support for the subsequent host vulnerability warning, and protected the security of cloud platform. Reduce the impact of attack traffic on the server.**

Keywords—Abnormal flow; Online detection; Neural network; Vulnerability warning

I. INTRODUCTION

The current network traffic data presents blowout growth, with large flow data and high flow rate[1][2]. Under the condition of ensuring a certain detection accuracy, it is an urgent problem to discover anomalies as soon as possible and alert the corresponding host of abnormal traffic related vulnerabilities[3][4]. Therefore, real-time online abnormal traffic detection is the key[5][6]. Traditional classification or statistical algorithms are time-consuming and have low timeliness requirements[7].

In the aspect of feature extraction, there are mainly two kinds of methods, linear extraction and nonlinear extraction[8]. Considering the useful information contained in the feature space and the data space, a feature extraction framework combining embedding learning and sparse regression is proposed[9][10]. By maximizing the internal activity similarity and minimizing the different activity similarity (MIMD) measurement, the best and most distinguishable feature set is selected from the original features extracted from the network packet sequence[11].

In the aspect of anomaly detection technology based on classification, the recursive neural network is applied to the problem of encrypted traffic classification, and the flow sequence network (FS-Net) is proposed[12]. FS-Net is an end-to-end classification model that learns representative features from raw streams and then classifies them in a multi-layer encoder-decoder framework[13].

In addition to classification technology, anomaly detection technology is also based on statistics and information theory. It relies on discrete time sliding window to continuously update feature space and uses incremental grid clustering to detect anomalies quickly, which can ensure low detection delay and good detection performance at the same time.

II. ALGORITHM DESCRIPTION

A. Flow feature extraction

First of all, through the analysis of network data frame, a data frame usually includes Ethernet link layer, IP network layer, transport layer and application layer four layers of encapsulation. The MAC address and protocol version contained in the link layer protocol header do not change in the same LAN. Therefore, it is not appropriate to use the link layer protocol header as the characteristic value. In this paper, IP packet header data and transport layer packet header data of network data frame are extracted as encrypted traffic characteristics.

The IP header, TCP header, and UDP header of data frame contain abundant feature information. In addition, a large number of features are time-dependent. TCP flows can be extracted through the timestamp of optional fields.

At the same time, considering the storage consumption of feature extraction and the timeliness of anomaly detection, the feature quantity threshold $\lambda$and the extraction time threshold φ for each stream are proposed in this paper. When the feature quantity extracted from one stream and the extractable quantity of the next packet are more than the feature threshold, the relevant features are extracted directly by filling zero instead of waiting for the next packet. When the extraction time exceeds the time threshold φ, the features of the stream are also output by zeroing.

B. Anomaly detection model

Convolutional Neural Networks (CNN), a deep learning algorithm that includes convolutional computation and deep feedforward neural networks, is widely used in image recognition and has a good effect. The main advantage of this method is that it can reduce the feature dimension of the data set while preserving the feature information.

A certain number of bytes of data header is extracted for each network traffic as the characteristics of the network flow. Since CNN has good detection performance in image recognition, this paper converted the pack-based traffic features into hexadecimal form to facilitate the input of the CNN model. Meanwhile, in order to avoid the features at the edge of the image being ignored, the mirror filling method was used for boundary filling.

In terms of model design, the structure of online anomaly detection system based on convolution is shown in Figure 1.

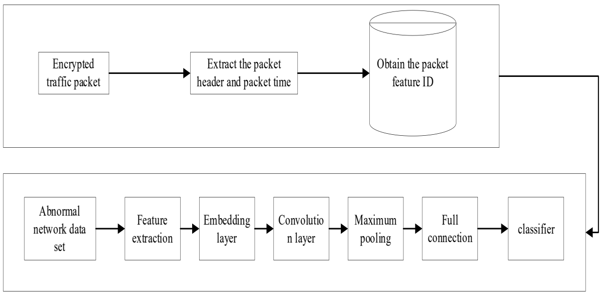

Fig. 1. Structure of online anomaly detection system based on convolution.

Firstly, the feature of encrypted traffic header is extracted online. The extraction idea designed in this paper is as follows: ID is extracted from the captured data packet quintuple, message with ID is sent to the message queue, and encrypted traffic header and time information is extracted and stored. The messages in the message queue are read concurrently to check whether the extracted network flow features meet the threshold requirements. If yes, the encrypted traffic features are output; otherwise, the features are stored and continued processing.

In the anomaly detection module, based on the classical LeNet-5, this paper still uses two convolutional layers, two pooling layers and two fully connected layers. First, the gray values of feature data are normalized. In the first layer, 32 5×5 convolution kernels are used to extract encrypted traffic features, and maximum pooling is applied to the pooling layer. The second layer is operated with 64 3×3 convolution kernels, followed by the maximum pooling operation at the pooling layer. All pooling layers use 2×2 Windows that slide two pixels at a time. Two fully connected layers are used for training. In this paper, ReLu is used as the activation function to reduce the overfitting problem of the model. Finally, the classification results of each node are synthesized by softMax classification function, and the actual classification results are obtained. softMax function takes the absolute value and sum of the classification probability of each node, and further normalizes the classification probability value of each node, so as to obtain the probability of each classification.

C. Algorithm implementation

The splitcap software is used to gather the original pcap packet data into a network flow and split into network flow pcap files in unidirectional flow. Secondly, python is used to extract encrypted traffic packet features that meet the conditions of byte number and time for each stream, and quintuple comparison is made with the stream label files provided by the data set to obtain real network stream label data. Table 1 describes the original data package of CICIDS-2017 data set.

TABLE I. COMPARISON OF AUC VALUES

file	Attack type	File size	Packet quantity
Monday	Benign (normal human activities)	10.62GB	11709971
Tuesday	Benign, FTP-Patator, SSH-Patator	11.05GB	11551954
Wednesday	Benign, DoS GoldenEye, DoS Hulk, DoS Slow httptest, DoS slowloris, Heartbleed	13.18GB	13788878
Thursday	Benign, Brute Force, SQL Injection, XSS, Infiltration	8.3GB	9322025
Friday	Benign, Bot, Port Scan, DDoS	8.84GB	9997874
Monday	Benign (normal human activities)	10.62GB	11709971

Based on the data set, the time and the size of the feature value in the feature extraction algorithm are evaluated. The CIC-IDS-2017 data set provides the Monday-WorkingHours.pcap file that only contains the original traffic packet of normal human activities. It is considered that it can represent the packet status of normal network traffic in the network environment. The file contains 11,709,971 network packets in total, which are extracted into 249,110 network flows by splitcap software using quintuple information.

III. EXPERIMENTAL VERIFICATION

The duration distribution of the flow in the file is calculated, and there is a large distribution interval with the duration of the flow[14][15]. In order to highlight the key data, the duration data of the flow is processed in this paper, and the processing method is shown in Equation 1.

$$t = \begin{cases} t,(t <= 1) \\ 1 + \log t,(1 < t < 10) \\ 2(t > 10) \end{cases} \qquad (1)$$

The duration is shown in Figure 2.

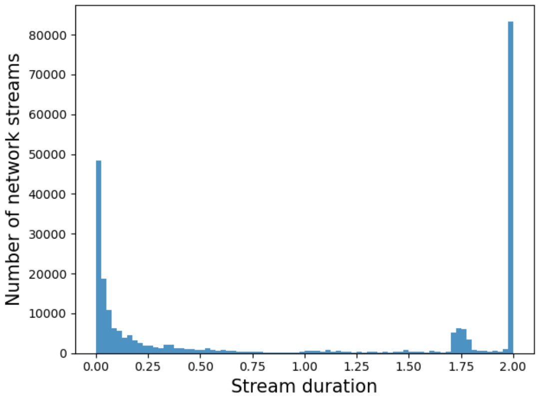

Fig. 2. Network traffic duration distribution.

It can be seen from the figure that the duration of streams is basically more than 1s. In order to ensure the effectiveness of anomaly detection, the extraction time threshold is set within 1s in this paper.

In this paper, the number of packets contained in each stream of TCP and UDP is analyzed respectively. Since the number of packets also has the problem of extreme value difference, the number of packets is processed and the traffic distribution is shown in Figure 3.

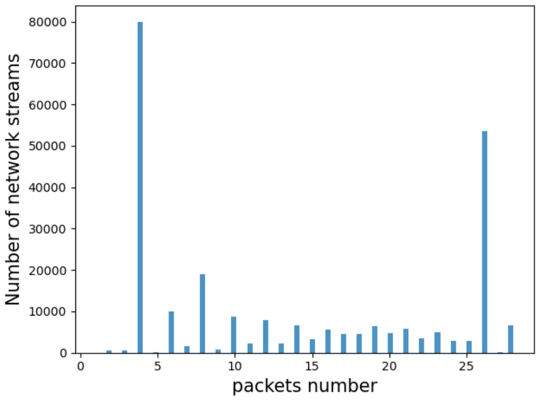

Fig. 3. Network traffic packet distribution.

At the same time, it can be seen that the number of traffic packets is basically less than 10. Considering that TCP establishes a connection with three handshakes, that is, 120 sub-sections, the extraction threshold of feature quantity is set as 289 bytes in this paper. Basically, four packet headers can be captured after the connection is established. At the same time, since this paper focuses on the study of online detection and pays more attention to the characteristics of network environment in the same time period, this paper separately conducts convolutional neural network experiments on data sets containing abnormal traffic.

Only a small number of network flow header messages can achieve a high accuracy of anomaly detection. The anomaly detection model based on convolutional neural network has relatively good detection performance. Therefore, it can be concluded that good anomaly detection performance can be achieved by using a small number of data header features in the early stage of network flow, which provides certain proof for the feasibility of online anomaly detection.

IV. CONCLUSIONS

In this paper, the data header itself is used as the encryption traffic feature. Firstly, the information contained in each layer of network packet header is analyzed, and through the analysis of common encryption traffic features, the encryption traffic feature extraction algorithm based on network layer, transport layer header and collection time stamp is proposed. Secondly, relying on the good performance of CNN in image recognition, this paper converts the characteristics of the packet header into gray image and conducts anomaly detection based on convolutional neural network. Simulation experiments prove that the datagram message information based on the early stage of network flow can perform network anomaly detection well. Through online detection of abnormal traffic, You can further send vulnerability warnings corresponding to related attacks to the host to protect the corresponding host security.

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